

contents

1. Completing tables of data from given information

Two puzzles where students have to use a set of clues to work out all the missing information in a table.

2. Interpreting tables of data

Students have to first work out the missing entries in a two way table and then read the information to decide if a series of statements are true or false. Includes easy fractions and percentages.

3. Calculating probabilities from tables of data

Again students must work out the missing entries in the table, then use that information to calculate the probability of various events.

two Way tableS 1

Use the clues to complete the table of data.

	Vanilla	Chocolate	Strawberry	Total
Girls				
Boys				
Total				

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The table shows the ice cream preferences of 44 children.

$\frac{3}{4}$ of the girls liked strawberry ice cream.

None of the girls liked Chocolate ice cream.

There were 4 more girls than boys.

6 boys liked vanilla ice cream.

$\frac{1}{4}$ of the children liked chocolate ice cream.

two Way tableS 2

Use the clues to complete the table of data.

	Maths	History	Art	Total
Year 6				
Year 7				
Year 8				
Total				

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The table shows which of three subjects 144 children liked the most.

$\frac{1}{3}$ of year 6 pupils liked Maths the most.

$\frac{1}{4}$ of the students were in year 6.

18 year 8 pupils liked Art the most.

There were equal numbers of students in year 7 and 8.

45 pupils liked History the most.

In year 6, 15 people liked Art the most.

Nobody in year 7 liked History.

In year 7 there were only 4 children who didn't list Maths as their favourite subject.

Answers

1

	Vanilla	Chocolate	Strawberry	Total
Girls	6	0	18	24
Boys	6	11	3	20
Total	12	11	21	44

Answers

2

	Maths	History	Art	Total
Year 6	12	9	15	36
Year 7	50	0	4	54
Year 8	0	36	18	54
Total	62	45	37	144

two Way tables : true or false?

Use the information in the table to work out which of these statements are true. There are 8 in total.

Survey Results: Children who play sports

	Plays a Team Sport	Plays an Individual Sport	Does not play sport	Total
Year 7		18		96
Year 8			35	94
Year 9	20	32		98
Total	60	100		288

A. $\frac{1}{3}$ of the children who completed the survey are in year 7.

D. $\frac{1}{3}$ of the children in year 7 play a team sport.

G. There are more children in year 7 than in year 9 who do not play a sport.

J. More than half of the children in year 8 do not play a sport.

B. Half of the children who play an individual sport are in year 8.

E. $\frac{1}{3}$ of the children who play a team sport are in year 9

H. There are equal numbers of children who play team sports and individual sports

K. Less than 10% of the children in year 8 play a team sport.

M. More children in year 7 play team sports than no sports.

C. 11 students in year 8 play a team sport.

F. 50% of the children who do not play sport are in year 8.

I. 128 children did not play any sport at all.

L. More than 50% of children in year 9 play some kind of sport.

N. More than half of the children who play team sports are in year 7.

True Statements: _____

two way
tables :
true or
false?

Answers.

Survey Results: Children who play sports

	Plays a Team Sport	Plays an Individual Sport	Does not play sport	Total
Year 7	31	18	47	96
Year 8	9	50	35	94
Year 9	20	32	46	98
Total	60	100	128	288

True Statements:

A B E G
I K L N

Two Way tables: Probability

The table shows some information about how 160 children get to school and what they normally have for breakfast.

Use the table to work out the following probabilities - the answers are jumbled up at the bottom.

	Hot breakfast	Cold breakfast	No breakfast	Total
Walks to School	0	6		24
Takes the Bus		14		32
By Car	20		10	
Total	28	94		160

A. The probability of a child arriving to school by car.

B. The probability of a child having no breakfast.

C. The probability that a child walks to school.

D. The probability that a child has some kind of breakfast.

E. The probability that a child who walks to school has a hot breakfast.

F. The probability that a child who takes the bus to school has a cold breakfast.

G. The probability that a child who doesn't eat breakfast gets to school by car.

H. The probability that a child who walks to school has breakfast.

I. The probability that a child who goes to school by car has breakfast.

J. The probability that a child who eats a hot breakfast takes the bus to school.

K. The probability of a child arriving to school by car and not having breakfast.

L. The probability of a child taking the bus to school and having a hot breakfast.

**Jumbled
Up
Answers**

$\frac{13}{20}$ $\frac{61}{80}$ $\frac{7}{16}$ $\frac{2}{7}$ $\frac{47}{52}$ $\frac{5}{52}$ $\frac{5}{19}$ **0** $\frac{1}{4}$ $\frac{1}{20}$ $\frac{3}{20}$ $\frac{19}{80}$

Record your answers here:

Statement	A	B	C	D	E	F	G	H	I	J	K	L
Probability												

two way tables : answers

The table shows some information about how 160 children get to school and what they normally have for breakfast.

Use the table to work out the following probabilities:

	Hot breakfast	Cold breakfast	No breakfast	Total
Walks to School	0	6	18	24
Takes the Bus	8	14	10	32
By Car	20	74	10	104
Total	28	94	38	160

Record your answers here as fractions in their simplest form:

Statement	A	B	C	D	E	F	G	H	I	J	K	L
Probability	$\frac{13}{20}$	$\frac{19}{80}$	$\frac{3}{20}$	$\frac{61}{80}$	0	$\frac{7}{16}$	$\frac{5}{19}$	$\frac{1}{4}$	$\frac{47}{52}$	$\frac{2}{7}$	$\frac{5}{52}$	$\frac{1}{20}$